

# BEE 208 — Question 5: Measurement Unit Conversion Menu

Control Statement: switch | Faculty of Engineering, ATU

## 1. ALGORITHM

Step 1: Start

Step 2: Declare variables choice (int), inputValue, result (double)

Step 3: Display the conversion menu (options 1 to 4)

Step 4: Prompt user to enter choice

Step 5: Read choice

Step 6: Use switch on choice

Step 7: Case 1 — read volts, compute millivolts = value  $\times$  1000, display result

Step 8: Case 2 — read amperes, compute milliamperes = value  $\times$  1000, display result

Step 9: Case 3 — read kilowatts, compute watts = value  $\times$  1000, display result

Step 10: Case 4 — read ohms, compute kilo-ohms = value / 1000, display result

Step 11: Default — display 'Invalid choice'

Step 12: End

## 2. PSEUDOCODE

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BEGIN

DECLARE choice AS INTEGER

DECLARE inputValue, result AS DOUBLE

OUTPUT menu (1.V→mV, 2.A→mA, 3.kW→W, 4.Ohm→kOhm)

INPUT choice

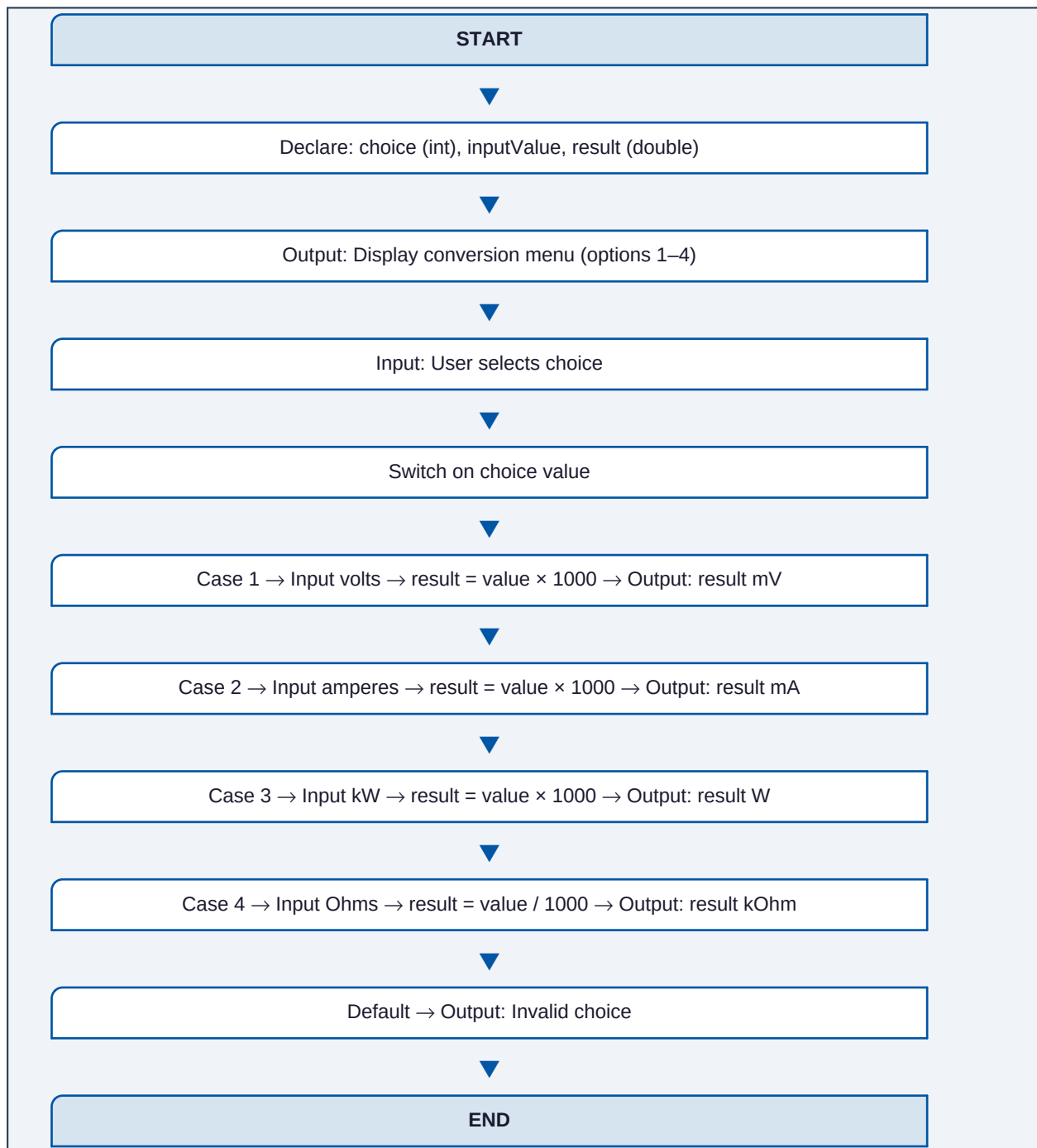
SWITCH choice

CASE 1: INPUT inputValue; result = inputValue * 1000; OUTPUT result, 'mV' BREAK
CASE 2: INPUT inputValue; result = inputValue * 1000; OUTPUT result, 'mA' BREAK
CASE 3: INPUT inputValue; result = inputValue * 1000; OUTPUT result, 'W' BREAK
CASE 4: INPUT inputValue; result = inputValue / 1000; OUTPUT result, 'kOhm' BREAK
DEFAULT: OUTPUT 'Invalid choice'

END SWITCH

END
```

## 3. FLOWCHART (Structured Diagram)



#### 4. TEST RESULTS

| Test Item | Value / Description      | Test Item       | Value / Description |
|-----------|--------------------------|-----------------|---------------------|
| Input     | choice=1, value=5 V      | Expected Output | 5000 mV             |
| Input     | choice=3, value=2 kW     | Expected Output | 2000 W              |
| Input     | choice=4, value=4700 Ohm | Expected Output | 4.7 kOhm            |